

One-Page Research Summary

Project: Explainable AI (XAI) with Conditional RAG and Research Agent

Objective: Develop an Explainable AI system that provides trustworthy, transparent, and evidence-backed responses using GPT, conditional Retrieval-Augmented Generation (RAG), and a dedicated Research Agent.

Architecture: The workflow begins with a Query Analyzer that determines whether document retrieval is required. Simple queries are answered directly by GPT, while document-based queries invoke the RAG pipeline. Supported inputs include PDF, Word, PowerPoint, Excel, images, audio, and video. Videos are processed using FFmpeg for audio extraction and Whisper for transcription.

Research Agent: After RAG retrieves relevant context, a Research Agent searches trusted scholarly sources (e.g., IEEE, ACM, Springer, Nature, arXiv, PubMed) to identify the most relevant research papers, summarize key findings, compare evidence, and provide refined context for GPT.

Explainability: The system records why RAG was executed or skipped, which tools were selected, which sources contributed to the answer, confidence scores, and the final reasoning path, improving transparency and user trust.

Expected Benefits: Reduced latency through conditional RAG, improved answer quality via research-backed evidence, efficient tool routing, scalable modular architecture, and enhanced explainability suitable for academic and industrial applications.